

# Maximizing Unlabeled Data Utility with Improved Representation Learning and Pseudo Labeling

Anonymous CVPR submission

Paper ID 17112

## Abstract

*An effective semi-supervised approach for representation learning involves two parallel networks - a trainable network and a momentum network. Two augmented versions of the input images go through these networks to predict their outputs and thereby learn the representation. However, this method often fails to maximize its full potential. We identify two major issues regarding data utilization: (1) the need for large batch sizes, which reduces efficiency in memory-constrained environments, and (2) the exclusion of unlabeled data during downstream tasks. We address these limitations with a two-step approach. First, we incorporate both original and augmented images into the trainable network and add a weighted loss term to enhance learning with the additional input. After pre-training, pseudo labels are generated in a meta-learning setup using two identical teacher-student networks, where the teacher network progressively outperforms the student by receiving feedback from it. Our approach, Representation Learning with Pseudo Labeling (RLPL), improves representation learning by achieving higher accuracy for half the batch size of existing methods. RLPL demonstrates an average performance gain of **6.12%** across the STL-10, CIFAR-10, and SVHN datasets, with individual improvements ranging from **2.7%** to **15.74%**.*

## 1. Introduction

While supervised machine learning algorithms thrive on labeled data, manually annotating this data can be both time-consuming and costly. In contrast, unlabeled data is often readily available and abundant. Although it lacks explicit labels, unlabeled data still holds valuable information that machine learning models can harness. Semi-supervised learning bridges this gap by extracting meaningful patterns from unlabeled data, compensating for the limited labeled datasets [26]. It achieves this by utilizing the unlabeled data to uncover latent representations—underlying struc-

tures and relationships that aren't directly observable. Semi-supervised learning can enhance model performance by incorporating these latent representations, even when labeled data is scarce [10, 28].

Contrastive learning [8, 12] has been proven effective in learning the data representation using the unlabeled sample. However, they require negative samples to avoid representational collapse [1] and hence were plagued by the problem of mining negative samples [39]. There are no apparent methods to obtain effective negative samples, and the presence of higher negative samples than positive ones creates an imbalance [24]. Subsequently, contrastive learning methods rely on a large batch size [57] to achieve good performance, leading to expensive computational needs.

Representation Learning (RL) methods, such as Bootstrap Your Own Latent (BYOL), SimSiam, and Barlow Twins solve the problem of negative samples by learning representation from the positive class alone. While these methods achieve impressive results with large batch sizes (typically 1024-4096), their performance degrades significantly with smaller batches. This degradation occurs because larger batches provide better gradient estimates and more diverse learning signals, which are crucial for stable representation learning. However, the high memory requirements of large batches make these methods impractical for resource-constrained environments.

Our analysis reveals that the performance drop in existing methods with smaller batches (8-128) stems from two key factors: (1) unstable gradient estimates leading to sub-optimal convergence, and (2) limited diversity in the learning signals within each batch. RLPL addresses these challenges through a novel approach that maximizes information utilization from each sample, enabling stable learning even with smaller batches. Through extensive experimentation across different batch sizes, we demonstrate that RLPL maintains consistent performance, showing only a marginal accuracy drop with reduced batch sizes.

Unlabeled data, typically unused in downstream tasks, can enhance performance when leveraged through a model with strong initialization. Pseudo-Labels (PL) [30] employ

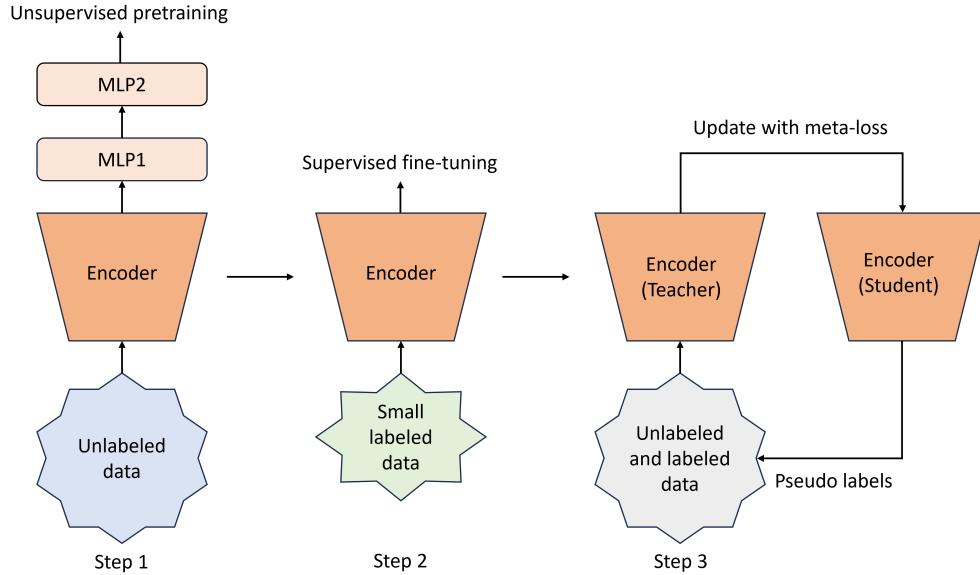


Figure 1. **RLPL flowchart.** RLPL has two levels of unlabeled data utilization. First, it employs unsupervised representation learning. Second, We obtain a highly accurate fine-tuned classifier using only a handful of label data. Lastly, through pseudo labelling using teacher-student networks, both initialized from superior supervised backbone, we improve the classifier further.

a confidence threshold to selectively include those unlabeled samples for which the classifier demonstrates high certainty in its predictions. RL offers an effective initialization, enabling improved model performance and confident pseudo-labeling. However, a notable challenge arises when incorrect pseudo labels are assigned, leading to a degradation in classifier accuracy due to learning from corrupted data. RLPL addresses this issue by refining the pseudo labels through a teacher-student network. This process enhances the teacher network, making it a better model to guide the student network, thereby generating more accurate pseudo labels and boosting overall performance.

Typical RL methods demand extensive computational resources, such as hundreds of TPUs over several days. Additionally, pseudo-labeling is another memory-heavy method as it uses two networks simultaneously. These requirements are challenging in scenarios where fast, efficient models are needed, like in autonomous vehicles or edge computing [16]. To address these limitations, we implement RLPL on a more manageable batch size and a shallower network, leveraging just two Nvidia 4090 GPUs. Our study shows that RLPL, even without the usage of pseudo labeling, achieves higher accuracy with a batch size of 64, outperforming BYOL with a 128 batch size. Remarkably, RLPL consistently outperforms other RLs with a ResNet18 classifier. RLPL yields average accuracy improvements of **2.7%**, **9.22%**, and **4.16%** that the closest contemporary method on the STL-10, CIFAR-10, and SVHN datasets, respectively.

Figure 1 illustrates our framework. In summary, our

work contains the following contributions:

- Our approach cleverly leverages most information from the inputs, leading to enhanced global representation learning. By making this strategic adjustment, our model achieves improved representations with a smaller batch size, ultimately achieving higher accuracy in the downstream tasks.
- During the downstream task, we utilize unlabeled samples a second time with pseudo labeling. Our approach uses the student network feedback with a meta-loss update to improve the noisy teacher [55] as the final classifier.
- RLPL outperforms other State-Of-The-Art (SOTA) RL algorithms on an equivalent but more compact architecture, making it more efficient and cost-effective for edge computing in resource-constrained environments.

## 2. Related Work

### 2.1. Representation Learning

Representation learning is a powerful technique for uncovering hidden features in unlabeled data, gaining popularity over time [17, 29, 43]. Initial methods, like auto-encoders [15, 59] and adversarial learning [11, 40, 42] from image samples, have their challenges. They often demand detailed inputs and hefty computational resources. Moreover, their outputs end up in pixel space instead of the more practical feature space, limiting their usefulness for downstream tasks.

A more contemporary approach to representation learning, contrastive learning [8, 49], has demonstrated greater effectiveness. Contrastive learning operates by bringing similar images together while pushing dissimilar ones apart. Supervised contrastive learning [25] executes in a fully supervised setting, leveraging label information to improve clustering in embedding space and achieve superior performance. The proposed Supervised Contrastive (SupCon) loss outperforms traditional cross-entropy loss on various datasets and demonstrates robustness to natural corruptions and stability to hyperparameter changes. On the other hand, Unsupervised Data Augmentation (UDA) [54] utilizes semi-supervised learning by enhancing the quality of noise through advanced data augmentation methods, significantly boosting model performance on various tasks. An influential algorithm in this domain, SimCLR [5], employs positive and negative image pairs for contrastive learning in a semi-supervised manner. SimCLR has exhibited remarkable performance, achieving SOTA accuracy on the ImageNet [9] dataset. However, SimCLR comes with significant computational expenses due to its reliance on large batch sizes and the incorporation of positive and negative pairs. These limitations render SimCLR impractical for many applications.

BYOL, SimSiam, Barlow Twin, and Momentum Contrast (MoCo) [18] are put forward as more efficient alternatives to SimCLR. A notable departure from SimCLR is eliminating the need for negative pairs during training. Instead, they leverage a pair of augmented views of the same image, streamlining the training process and reducing computational overhead. This design choice positions these RL algorithms as computationally more efficient options than SimCLR. SimCLRv2 [6] uses a big pretrained model for a second stage data utilization. In contrast, our approach eliminates the reliance on large models by maintaining a consistent model architecture across all training phases.

## 2.2. Pseudo Labeling

In semi-supervised learning, pseudo-labeling is used on a highly accurate classifier to label and learn from it. This approach has proven effective in various applications like object detection [32, 56], image classification [48, 52], and speech recognition [34]. If pseudo labels are inaccurately assigned, classifiers may learn from erroneous labels, diminishing their performance [2].

Many pseudo-labeling methods focus on enhancing the network through consistency loss [58], moving away from the traditional approach of retraining the model with pseudo labels [23]. Consistency-based methods ensure stable model predictions across various augmented views of the same input [41]. However, consistency loss typically updates the network with a fraction of the cross-entropy loss [50], reducing its impact. This diminished emphasis on consistency

loss can hinder the network’s optimal update, as it does not fully leverage retraining with pseudo labels [54].

MPL addresses the issue of incorrect pseudo-labeling by applying a meta-learning [31] approach. In MPL, the teacher uses its softmax predictions on unlabeled data to instruct the student. These predictions, commonly called soft labels, are a well-established concept in the literature on knowledge distillation [7, 38]. This additional adaptation empowers the teacher to generate more accurate PLs for instructing the student network. MPL employs the same bi-level optimization technique from meta-learning [13, 35, 46] to iteratively update the teacher-student framework.

Notably, MPL progressively enhances the student network, while our training focuses more on improving the teacher network. This is because the teacher network receives more frequent feedback and updates - first through feedback from the student and then from the meta loss update. Additionally, introducing a controlled noise in either the teacher or student network is crucial, especially when the two networks are similar. This noise injection boosts the learning process’s robustness and generalization [33, 51].

## 3. Proposed Methodology: RLPL

### 3.1. Improved Representation Learning

In RLPL, the trainable and momentum networks use identical encoders for image embeddings. These embeddings are projected to a lower-dimensional space using a Multi-Layer Perceptron (MLP) in each network. The momentum network stops here, producing outputs with a dimension of 256. In contrast, the trainable network includes an additional prediction head, consisting of another MLP layer with the same dimension.

RLPL learns a representation  $y_\theta$  for downstream tasks using encoder  $f_\theta$ , a projector  $g_\theta$ , and a predictor  $q_\theta$ . The momentum network,  $f_\xi$ , with a projector  $g_\xi$ , mirrors the trainable network’s architecture but updates as an exponential moving average with decay rate,  $\gamma \in [0, 1]$ :

$$\xi \leftarrow \gamma\xi + (1 - \gamma)\theta. \quad (1)$$

For an input image  $x$ , we create two augmented views  $v$  and  $v'$ . The trainable network computes  $y_\theta \triangleq f_\theta(v)$  and  $z_\theta \triangleq g_\theta(y_\theta)$ , while the momentum network outputs  $y'_\xi \triangleq f_\xi(v')$  and  $z'_\xi \triangleq g_\xi(y'_\xi)$ .

We then predict  $z'_\xi$  from  $z_\theta$  using  $q_\theta(z_\theta)$ , and  $l_2$ -normalize both  $q_\theta(z_\theta)$  and  $z'_\xi$  to  $\bar{q}_\theta(z_\theta) \triangleq \frac{q_\theta(z_\theta)}{\|q_\theta(z_\theta)\|_2}$  and  $\bar{z}'_\xi \triangleq \frac{z'_\xi}{\|z'_\xi\|_2}$ . The loss function is the mean squared error between the normalized prediction and momentum projection (as shown in Figure 2a):

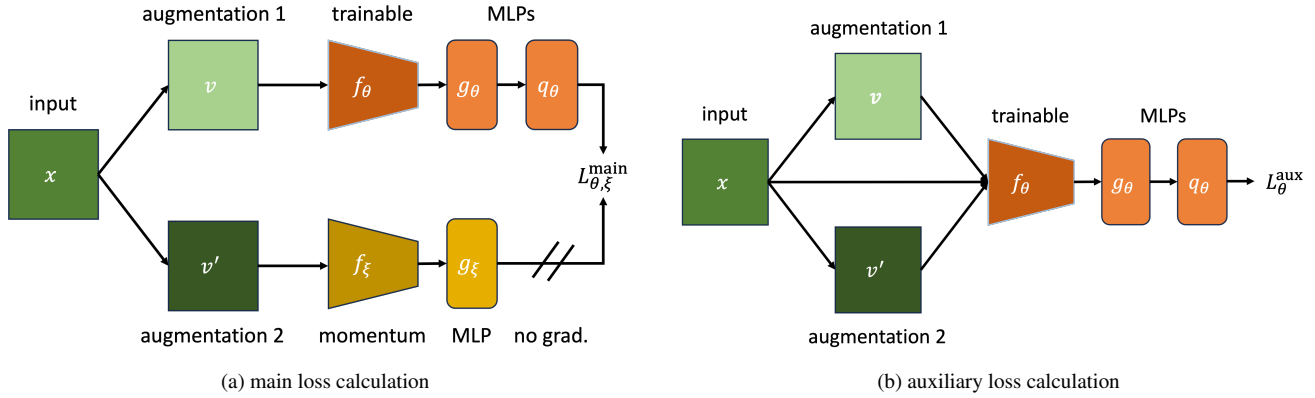


Figure 2. Representation learning with RLPL: (a) The augmented inputs are processed through both networks to compute the main loss. (b) The original and augmented inputs are fed into the trainable network to compute the auxiliary loss.

$$L_{\theta,\xi} \triangleq \|\bar{q}_\theta(z_\theta) - \bar{z}'_\xi\|_2^2 = 2 - 2 \cdot \frac{\langle q_\theta(z_\theta), z'_\xi \rangle}{\|q_\theta(z_\theta)\|_2 \cdot \|z'_\xi\|_2}. \quad (2)$$

To symmetrize the loss, we feed  $v'$  to the trainable network and  $v$  to the momentum network, computing  $\tilde{L}_{\theta,\xi}$ . The combined loss for optimization is:

$$L_{\theta,\xi}^{\text{main}} = L_{\theta,\xi} + \tilde{L}_{\theta,\xi}. \quad (3)$$

To calculate the auxiliary loss, the original image goes through the online network as an additional input (as shown in Figure 2b). For original input  $x$  and augmented views  $v, v'$ , we obtain outputs  $q_\theta(z_\theta^x) \triangleq q_\theta(g_\theta(f_\theta(x)))$ ,  $q_\theta(z_\theta^v) \triangleq q_\theta(g_\theta(f_\theta(v)))$ , and  $q_\theta(z_\theta^{v'}) \triangleq q_\theta(g_\theta(f_\theta(v')))$ , respectively. We design an auxiliary loss that allows gradient flow during backpropagation. This auxiliary loss is combined with the main loss, weighted by a factor  $\lambda$ . The momentum network stays the same, but the trainable network affects its update parameters.

We calculate the mean of two output embeddings,  $\bar{q}_\theta(z_\theta^{vv'})$ , from the prediction head.

$$\bar{q}_\theta(z_\theta^{vv'}) = \frac{1}{2} (\bar{q}_\theta(z_\theta^v) + \bar{q}_\theta(z_\theta^{v'})) \quad (4)$$

where  $\bar{q}_\theta(z_\theta^v)$  and  $\bar{q}_\theta(z_\theta^{v'})$  are the batch-mean of their respective embeddings,  $q_\theta(z_\theta^v)$  and  $q_\theta(z_\theta^{v'})$ .

Now, we denote the auxiliary loss as  $L_{\theta}^{\text{aux}}$  and it is expressed as,

$$L_{\theta}^{\text{aux}} = \|\bar{q}_\theta(z_\theta^x) - \bar{q}_\theta(z_\theta^{vv'})\|_2^2 \quad (5)$$

where  $\bar{q}_\theta(z_\theta^x)$  is the batch-mean of  $q_\theta(z_\theta^x)$ .

Next, it is added to the main loss with a weight,  $\lambda$ , to obtain the total loss.

$$L_{\theta,\xi} = L_{\theta,\xi}^{\text{main}} + \lambda L_{\theta}^{\text{aux}} \quad (6)$$

Optimization updates the trainable network parameters  $\theta$  with a learning rate  $\eta$ , while the momentum network parameters are updated as follows:

$$\theta \leftarrow \text{optimizer}(\theta, \nabla_{\theta} L_{\theta,\xi}, \eta), \quad (7)$$

$$\xi \leftarrow \gamma \xi + (1 - \gamma) \theta. \quad (8)$$

After training, only the encoder  $f_\theta$  is returned for the downstream tasks. The pseudo-code for these steps is provided in Algorithm 1.

### 3.2. Effective Pseudo Labeling

For effective pseudo-labeling, let,  $f_{\theta_T}$  and  $f_{\theta_S}$  denote teacher and student networks initialized with parameters  $\theta_T, \theta_S \leftarrow \theta$  from the pretrained encoder. Given a batch of labeled samples  $(x_l, y_l) \in \mathcal{D}_l$  and unlabeled samples  $x_u \in \mathcal{D}_u$ , we formulate an iterative pseudo-labeling framework.

For each iteration, we first compute the student's initial loss on labeled data:

$$L_{s1} = \mathbb{E}_{(x_l, y_l) \sim \mathcal{D}_l} [\text{CE}(f_{\theta_S}(x_l), y_l)] \quad (9)$$

For unlabeled data  $x_u$ , the teacher network generates soft pseudo-labels through weak augmentation  $\mathcal{A}$ :

$$\hat{y}_u = f_{\theta_T}(\mathcal{A}(x_u)) \quad (10)$$

To ensure quality of pseudo-labels, we filter predictions based on confidence threshold  $\tau$ :

$$\mathcal{M}_u = \{(x_u, \hat{y}_u) | \max(f_{\theta_T}(\mathcal{A}(x_u))) \geq \tau\} \quad (11)$$

**Algorithm 1** Representation learning with RLPL

---

```

# f_t: trainable net (backbone + projector + predictor)
# f_m: momentum net (backbone + projector)
# lambda: weight of auxiliary loss
# gma: momentum update rate

# load a minibatch x with n samples
for x in loader:
    v1, v2 = augment(x) # two random augmentations

    # compute embeddings
    t, t1, t2 = f_t(x), f_t(v1), f_t(v2), # N x D

    with torch.no_grad(): # stop the gradient
        m1, m2 = f_m(v1), f_m(v2) # N x D

    # calculate the losses
    loss_main = (loss1(t1, m2) + loss1(t2, m1)).mean()
    loss_aux = loss2(t, t1, t2)
    total_loss = loss_main + lambda * loss_aux

    # optimization step
    total_loss.backward()
    update(f_t.params)

    # momentum network update
    f_m.params = gma*f_m.params+(1-gma)*f_t.params

def loss1(a, b): # main loss function
    a = normalize(a, dim=-1)
    b = normalize(b, dim=-1)
    loss = 2 - 2 * (a * b).sum(dim=-1)
    return loss.mean()
def loss2(a, b, c): # auxiliary loss function
    avr_emb = mean(b, c, dim=-1)
    loss = (a - avr_emb) ** 2
    return loss.mean()

```

---

282 The teacher then learns from the filtered pseudo-labeled  
283 data:

$$L_u = \mathbb{E}_{(x_u, \hat{y}_u) \sim \mathcal{M}_u} [\text{CE}(f_{\theta_T}(x_u), \hat{y}_u)] \quad (12)$$

285 After updating the student parameters using  $L_u$ , we mea-  
286 sure its new performance:

$$L_{s2} = \mathbb{E}_{(x_l, y_l) \sim \mathcal{D}_l} [\text{CE}(f_{\theta_S}(x_l), y_l)] \quad (13)$$

288 The effectiveness of pseudo-labeling is captured by the  
289 performance change:

$$\Delta L_s = L_{s2} - L_{s1} \quad (14)$$

290 We incorporate this feedback through a meta-loss:  
291

$$L_{\text{meta}} = \Delta L_s \cdot L_u \quad (15)$$

The final objective for updating the teacher combines su-  
pervised learning on labeled data with the meta-loss feed-  
back:

$$L_{\text{total}} = \underbrace{\mathbb{E}_{(x_l, y_l) \sim \mathcal{D}_l} [\text{CE}(f_{\theta_T}(x_l), y_l)]}_{\text{supervised loss}} + \underbrace{L_{\text{meta}}}_{\text{feedback}} \quad (16)$$

This formulation creates a feedback loop where the  
teacher’s pseudo-labeling strategy is dynamically adjusted  
based on the student’s learning outcomes. When  $\Delta L_s < 0$ ,  
indicating improved student performance, the teacher is en-  
couraged to maintain its current strategy. Conversely, when  
 $\Delta L_s > 0$ , suggesting degraded performance, the teacher  
receives a stronger gradient signal to modify its approach.

The optimization proceeds iteratively:

$$\theta_S \leftarrow \theta_S - \eta_S \nabla_{\theta_S} L_u \quad (17)$$

$$\theta_T \leftarrow \theta_T - \eta_T \nabla_{\theta_T} L_{\text{total}} \quad (18)$$

where  $\eta_S$  and  $\eta_T$  are learning rates for student and  
teacher networks respectively. After training convergence,  
only the teacher network is retained for deployment.

The pseudo-code for our pseudo labeling approach is  
provided in Algorithm 2.

## 4. Experiments

### 4.1. Datasets

In our experiments, we applied STL-10 [45], CIFAR-10  
[20] and SVHN [44] datasets for unlabeled RL training. The  
STL-10 dataset is specially designed for unsupervised fea-  
ture learning as it contains 13,000 labeled images across 10  
classes and 100,000 unlabeled images. While we used the  
entire training set of CIFAR-10 and SVHN datasets for RL,  
we only used 250, 1,000, and 4,000 labeled training sam-  
ples in downstream classifications. Additional experiments  
were conducted on 6 other image datasets [19, 21, 22, 27,  
36, 53] with different class numbers.

### 4.2. Implementation Details

**Image augmentations:** We adopt an augmentation pipeline  
similar to SimCLR and BYOL, implemented in PyTorch  
[37]. For geometric transformations, we apply “Random-  
ResizedCrop” with a scaling factor between [0.2, 1.0] and  
“RandomHorizontalFlip” to introduce random horizontal  
flips. Color augmentations include “ColorJitter”, which  
modifies brightness, contrast, and saturation by a maximum  
factor of 0.4, and hue by 0.1, applied with a probability of  
0.3. We include “RandomGrayscale” to convert images to



**Algorithm 2** Pseudo labeling with RLPL

```

# f_t: pretrained encoder
# n_t, n_s: teacher and student networks
# tau: pseudo labels threshold

# initialize teacher and student networks
n_t = f_t.state_dict()
n_s = f_t.state_dict()

# load a minibatch of labeled and unlabeled samples
for x, y, x_u in loader:
    with torch.no_grad(): # stop the gradient
        loss_s1 = cross_entropy(n_s(x), y)
    y_u = n_t(x_u)
    # filtering unlabeled samples based on threshold
    x_u, y_u = x_u [y_u ≥ tau], y_u [y_u ≥ tau]

    y_ut = n_t(augment(x_u)) # weak augmentation
    loss_t = cross_entropy(y_ut, y_u)
    loss_t.backward()
    update(n_s.params) # student network update

    with torch.no_grad(): # stop the gradient
        loss_s2 = cross_entropy(n_s(x), y)
    change = loss_s2 - loss_s1
    meta_loss = change * loss_t # meta-loss computation

    loss = cross_entropy(n_t(x), y)
    total_loss = loss + meta_loss # total loss
    total_loss.backward()
    update(n_t.params) # teacher network update

```

grayscale with a probability of 0.2. For blurring, “Gaussian-Blur” is applied, using a Gaussian kernel with a standard deviation sampled from  $[1.0, 2.0]$ , with a 0.2 probability. Finally, we normalize the input images using predefined mean and standard deviation values.

Our experiments reveal that image resolution significantly affects accuracy; thus, we resize all images to  $256 \times 256$ . In the PL phase, weak augmentations are applied to the noisy teacher model, involving random horizontal and vertical flips, each with a probability of 0.25.

**Architecture:** We use ResNet18 [3], a more computationally feasible network, designed for GPU computations. This ResNet18 architecture remains consistent across all settings, including representation learning, and pseudo labeling. The MLPs, projection, and prediction layers adhere to the parameters established by BYOL. In PL, the teacher and student networks are initialized from the ResNet18, pre-trained via RL.

**Optimization:** The RL framework was optimized using the Adam optimizer [60], whereas PL employed the Stochastic

Gradient Descent (SGD) optimizer [14], both with a learning rate set to 0.0003. The auxiliary loss is weighted at 0.5. RL underwent training for 50 epochs with a batch size of 128, while PL was trained over 150 epochs with a batch size of 16.

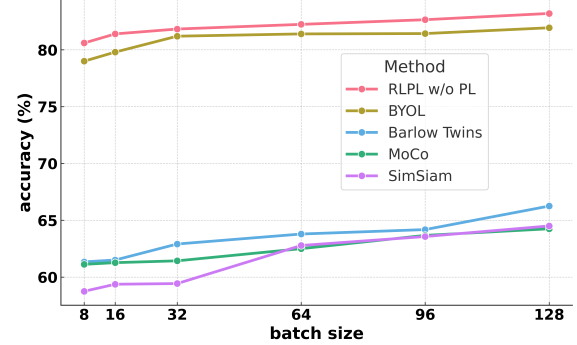


Figure 3. Effect of batch size on model performance (evaluated on STL-10).

| Methods             | Test Accuracy (%) |
|---------------------|-------------------|
| Baseline (ResNet18) | 53.16             |
| FixMatch [47]       | 37.28             |
| MixMatch [4]        | 71.60             |
| MoCo                | 66.25             |
| SimSiam             | 64.50             |
| Barlow Twins        | 64.25             |
| BYOL                | 81.94             |
| <b>RLPL (Ours)</b>  | <b>84.64</b>      |

Table 1. Performance of different methods on STL-10 dataset.

## 5. Results and Discussion

Our evaluation methods primarily comprise the baseline model, BYOL, SimSiam, MoCo, Barlow Twins, RLPL, and two additional semi-supervised algorithms, FixMatch [47], and MixMatch [4] on the considered datasets. In all our experiments, we re-implemented each algorithm on a ResNet18 backbone with identical hyperparameters to ensure a fair comparison.

### 5.1. Effect of Batch Size

To illustrate the effectiveness of RLPL with reduced batch sizes, we present the results in Figure 3. We started with a batch size of 8 for RLPL. Subsequently, we gradually increased the batch sizes for RLPL and others to evaluate their performances. To have a fair comparison, we applied the classification tasks only on the learned representation of RL, without any use of PL. As anticipated, the accuracy diminished with smaller batch sizes. Remarkably, at

| Method       | CIFAR-10     |              |              | SVHN         |              |              | Average      |
|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|              | 250 labels   | 1000 labels  | 4000 labels  | 250 labels   | 1000 labels  | 4000 labels  |              |
| ResNet18     | 33.90        | 43.71        | 58.47        | 41.20        | 84.58        | 87.60        | 58.24        |
| MoCO         | 34.50        | 46.75        | 61.88        | 30.75        | 83.25        | 89.12        | 57.37        |
| SimSiam      | 23.88        | 29.50        | 48.88        | 14.62        | 51.25        | 67.38        | 39.92        |
| Barlow Twins | 35.88        | 47.75        | 63.75        | 21.88        | 79.01        | 88.25        | 56.42        |
| BYOL         | 33.19        | 46.92        | 59.29        | 67.00        | 85.60        | 88.20        | 63.70        |
| RLPL (ours)  | <b>48.93</b> | <b>52.45</b> | <b>65.69</b> | <b>72.12</b> | <b>88.39</b> | <b>92.78</b> | <b>70.39</b> |

Table 2. Test accuracy (%) of different methods on varying numbers of training samples.

| Method            | STL-10       | CIFAR-10     |              |              | SVHN         |              |              | Average      |
|-------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                   |              | 250 labels   | 1000 labels  | 4000 labels  | 250 labels   | 1000 labels  | 4000 labels  |              |
| PL (random init.) | 53.34        | 30.49        | 40.57        | 55.90        | 11.19        | 17.62        | 75.62        | 40.39        |
| RLPL w/o PL       | 83.20        | 34.54        | 48.48        | 59.60        | 71.25        | 85.54        | 89.00        | 67.66        |
| RLPL (ours)       | <b>84.64</b> | <b>48.93</b> | <b>52.45</b> | <b>65.69</b> | <b>72.12</b> | <b>88.39</b> | <b>92.78</b> | <b>72.14</b> |

Table 3. Test accuracy (%) for ablation on methods.

a batch size of 64, RLPL attained an accuracy of 82.24%, surpassing BYOL’s accuracy of 81.94% with a batch size of 128. All other methods, including Barlow Twins, MoCo, and SimSiam, achieve significantly lower accuracy in the small-scale implementation, with their performance further declining as batch size decreases. Although all methods exhibited a decline in accuracy with further reductions in batch sizes, RLPL consistently outperformed BYOL and all the others.

## 5.2. Classification Accuracy

Table 1 reports the classification accuracy on the STL-10 dataset. The ResNet18 classifier demonstrates challenges in reaching acceptable accuracy levels. For a fair comparison, all methods, including FixMatch and MixMatch, were implemented with a ResNet18 backbone and an identical batch size.

The results reveal that FixMatch struggles to achieve competitive accuracy, likely due to the limitations of pseudo-labeling techniques, which often require robust initialization and a deep backbone to perform effectively. While MixMatch shows considerable improvement, it still falls short of the performance attained by BYOL and RLPL. MoCo, SimSiam, and Barlow Twins achieve almost similar but lower accuracy. Notably, BYOL achieves an accuracy of 81.94%, whereas RLPL surpasses this with an accuracy of 84.64%, obtaining a 2.7% improvement over BYOL.

Table 2 compares ResNet18, the four primary methods central to this study, and RLPL across varying training samples. For both datasets, all methods showed progressively higher accuracies with increasing sample sizes. We observe that our method uniquely achieves acceptable accuracy on CIFAR-10 with only 250 labels, whereas other

methods struggle under the same conditions. For SVHN with 250 labels, only BYOL and RLPL reach satisfactory accuracy, while other methods perform notably worse. Notably, RLPL consistently achieved the highest accuracy across all datasets and under all varying configurations, outperforming other methods in each case.

We include a column displaying the average test accuracy across all setups to provide a comprehensive view of the relative performance of each method. Only BYOL comes close to RLPL’s performance; however, it still lags by a margin of 6.69%. Nevertheless, RLPL consistently outperformed BYOL across all scenarios, particularly with smaller sample sizes, with improvement margins ranging from 2.79% to 15.74%.

## 5.3. Transfer to Other Classification Tasks

To assess RLPL’s generalization capabilities, we evaluated it on a range of classification tasks across diverse datasets, as summarized in Table 4. This table compares RLPL’s performance with baseline and other methods after pretraining on the STL-10 dataset, followed by the downstream tasks. For a fair comparison, we applied the pretrained RLPL model directly, without using PLs in the classification tasks. This approach highlights RLPL’s standalone performance; notably, integrating PL strategies could likely enhance accuracy beyond the presented results.

RLPL achieves the highest accuracy in nearly all cases, except for MoCo on the Labeled Faces in the Wild (LFW) dataset and BYOL on the EuroSAT dataset. RLPL obtains an average accuracy of 81.51%, substantially outperforming methods like SimSiam and Barlow Twins. Moreover, RLPL obtains 2.84% more accuracy than the closest best method, BYOL, across all the transfer learning tasks. This improve-

| Method       | CIFAR100     | FashionMNIST | LFW          | EuroSAT      | OxfordFlowers102 | Imagenette   | Average      |
|--------------|--------------|--------------|--------------|--------------|------------------|--------------|--------------|
| ResNet18     | 58.09        | 94.55        | 85.03        | 93.62        | 38.48            | 80.66        | 75.07        |
| MoCo         | 62.50        | 90.54        | <b>93.21</b> | 95.00        | 35.00            | 79.88        | 76.69        |
| SimSiam      | 33.88        | 92.63        | 80.41        | 79.38        | 26.12            | 76.82        | 64.54        |
| Barlow Twins | 58.75        | 94.38        | 89.19        | 94.62        | 27.88            | 80.38        | 74.87        |
| BYOL         | 63.61        | 93.62        | 80.73        | <b>96.28</b> | 49.64            | 86.13        | 78.67        |
| RLPL w/o PL  | <b>68.08</b> | <b>96.01</b> | 89.33        | 96.03        | <b>51.67</b>     | <b>87.94</b> | <b>81.51</b> |

Table 4. Performance comparison (% test accuracy) of methods on various datasets.

ment highlights RLPL’s transferability on unseen classification tasks.

## 6. Ablation Study

### 6.1. Framework Ablation

We conduct extensive ablation studies to evaluate the effectiveness of our method against the PL algorithm with random initialization and RLPL without PLs. Results in Table 3 demonstrate the consistent superiority of RLPL across diverse datasets and label regimes. The PL with random initialization exhibits limited performance, particularly struggling with low-label scenarios (e.g., 30.49% and 11.19% accuracy with 250 labels), highlighting the importance of proper initialization for PL’s reliability.

RLPL without pseudo labeling shows substantial improvements over the baseline, achieving notable gains on STL-10 (83.20% vs 53.34%) and SVHN (71.25% vs 11.19% for 250 labels), validating the effectiveness of our representation learning strategy. The full RLPL framework, integrating both representation learning and pseudo labeling, consistently achieves the highest performance across all configurations, with particularly strong improvements in low-label settings (e.g., 48.93% on CIFAR-10 with 250 labels, an 18.44% improvement). This comprehensive performance advantage is reflected in the average accuracy across all settings, demonstrating RLPL’s robust capability, especially in label-scarce scenarios where traditional approaches typically falter.

### 6.2. Ablation on Noisy Teacher

| Method            | Accuracy (%)     |
|-------------------|------------------|
| Noisy Student     | 84.54 $\pm$ 0.05 |
| Noiseless Teacher | 84.59 $\pm$ 0.02 |
| Noisy Teacher     | 84.63 $\pm$ 0.02 |

Table 5. Input types ablation for pseudo labeling.

We investigate the impact of noisy samples (weakly augmented) in both teacher and student networks. Table 5 presents the average performance of five different runs for three configurations: the noisy student, the noiseless

teacher, and our proposed noisy teacher method. While the differences are subtle, they reveal interesting patterns in model behavior. The noisy student achieves 84.54% accuracy with a standard deviation of 0.05, suggesting some variability in performance. When we remove noisy samples from the teacher network, we observe a marginal improvement to 84.59% with reduced variance. Our proposed approach achieves the highest accuracy at 84.63% with consistently low variance. These results suggest that weak augmentation in the teacher network inputs can provide additional regularization benefits.

## 7. Parameter Studies

### 7.1. Weight of the Auxiliary Loss

We modify the weight of auxiliary loss,  $\lambda$ , from 0.25 to 1.0 with an increment of 0.25. We observe that there is a strong effect of the weight on model performance. For weights other than 0.50, the test accuracy declines as we move further from this value. The outputs are shown in Figure 4a.

### 7.2. Threshold for Pseudo Labels

We vary the threshold,  $\tau$ , from 0.85 to 1.0 with an increment of 0.05. We observe that the highest accuracy is obtained at 0.95, and it keeps decreasing for a lower threshold value. Again, the accuracy decreases for a threshold value of 1.0. For any threshold other than 0.95, the accuracy drops below 84.0%. The outputs are shown in Figure 4b.

## 8. Conclusion

We present RLPL, a novel semi-supervised learning framework designed to maximize the utility of unlabeled data, addressing critical limitations in existing methods such as the reliance on large batch sizes and underutilization of unlabeled data in downstream tasks. RLPL demonstrates consistent improvements over SOTA methods, achieving high accuracy with smaller batch sizes, which enhances efficiency for memory-constrained environments. Our experiments show that RLPL yields substantial performance gains across multiple datasets, with notable advantages in settings with limited labeled data. In addition to accuracy improvements, RLPL demonstrates exceptional transferability, achieving strong performance across a diverse range of



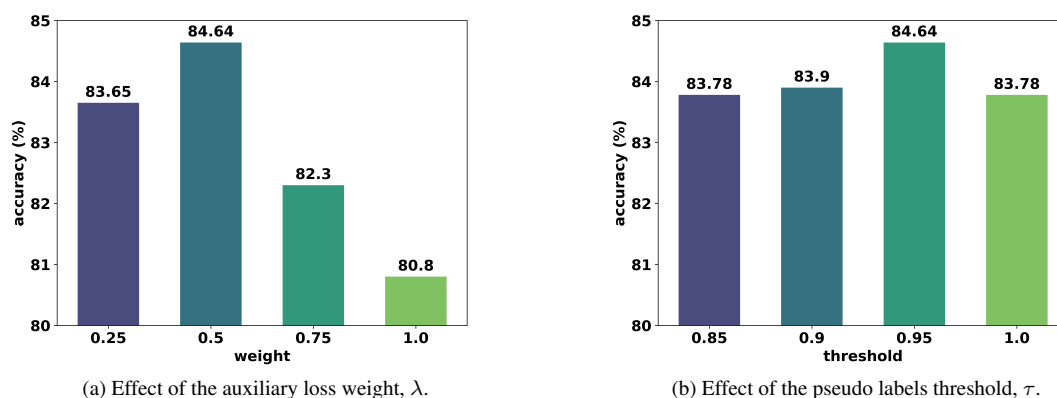


Figure 4. Effects of different hyperparameters on model performance.

downstream classification tasks without needing pseudo labels for adaptation. By offering enhanced data efficiency, robustness, and generalization, RLPL is ideally suited for a variety of applications that demand memory efficiency and cross-task adaptability, marking a significant advancement in semi-supervised learning.

## References

- [1] Armen Aghajanyan, Akshat Shrivastava, Anchit Gupta, Naman Goyal, Luke Zettlemoyer, and Sonal Gupta. Better fine-tuning by reducing representational collapse. *arXiv preprint arXiv:2008.03156*, 2020. 1
- [2] Eric Arazo, Diego Ortego, Paul Albert, Noel E O'Connor, and Kevin McGuinness. Pseudo-labeling and confirmation bias in deep semi-supervised learning. In *2020 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2020. 3
- [3] Swarnambiga Ayyachamy, Varghese Alex, Mahendra Khened, and Ganapathy Krishnamurthi. Medical image retrieval using resnet-18. In *Medical imaging 2019: imaging informatics for healthcare, research, and applications*, pages 233–241. SPIE, 2019. 6
- [4] David Berthelot, Nicholas Carlini, Ian Goodfellow, Nicolas Papernot, Avital Oliver, and Colin A Raffel. Mixmatch: A holistic approach to semi-supervised learning. *Advances in neural information processing systems*, 32, 2019. 6
- [5] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PMLR, 2020. 3
- [6] Ting Chen, Simon Kornblith, Kevin Swersky, Mohammad Norouzi, and Geoffrey E Hinton. Big self-supervised models are strong semi-supervised learners. *Advances in neural information processing systems*, 33:22243–22255, 2020. 3
- [7] Jang Hyun Cho and Bharath Hariharan. On the efficacy of knowledge distillation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 4794–4802, 2019. 3
- [8] Ching-Yao Chuang, Joshua Robinson, Yen-Chen Lin, Antonio Torralba, and Stefanie Jegelka. Debaised contrastive learning. *Advances in neural information processing systems*, 33:8765–8775, 2020. 1, 3
- [9] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009. 3
- [10] Benjamin Devillers, Léopold Maytié, and Rufin VanRullen. Semi-supervised multimodal representation learning through a global workspace. *IEEE Transactions on Neural Networks and Learning Systems*, 2024. 1
- [11] Jeff Donahue and Karen Simonyan. Large scale adversarial representation learning. *Advances in neural information processing systems*, 32, 2019. 2
- [12] Rizhao Fan, Matteo Poggi, and Stefano Mattoccia. Contrastive learning for depth prediction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 3226–3237, 2023. 1
- [13] Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International conference on machine learning*, pages 1126–1135. PMLR, 2017. 3
- [14] Robert Mansel Gower, Nicolas Loizou, Xun Qian, Alibek Sailanbayev, Egor Shulgin, and Peter Richtárik. Sgd: General analysis and improved rates. In *International conference on machine learning*, pages 5200–5209. PMLR, 2019. 6
- [15] Praseon Goyal, Zhiting Hu, Xiaodan Liang, Chenyu Wang, and Eric P Xing. Nonparametric variational auto-encoders for hierarchical representation learning. In *Proceedings of the IEEE International Conference on Computer Vision*, pages 5094–5102, 2017. 2
- [16] Xiaojun Guo, Yifei Wang, Zeming Wei, and Yisen Wang. Architecture matters: Uncovering implicit mechanisms in graph contrastive learning. *Advances in Neural Information Processing Systems*, 36:28585–28610, 2023. 2
- [17] William L Hamilton, Rex Ying, and Jure Leskovec. Representation learning on graphs: Methods and applications. *arXiv preprint arXiv:1709.05584*, 2017. 2

- [18] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. *arXiv e-prints*, art. *arXiv preprint arXiv:1911.05722*, 2019. 3
- [19] Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7):2217–2226, 2019. 5
- [20] Tien Ho-Phuoc. Cifar10 to compare visual recognition performance between deep neural networks and humans. *arXiv preprint arXiv:1811.07270*, 2018. 5
- [21] Jeremy Howard. Imagenette: A smaller dataset for image classification experiments. <https://github.com/fastai/imagenette>, 2019. 5
- [22] Gary B. Huang, Manu Ramesh, Tamara Berg, and Erik Learned-Miller. Labeled faces in the wild: A database for studying face recognition in unconstrained environments. Technical Report 07-49, University of Massachusetts, Amherst, 2007. 5
- [23] Ahmet Iscen, Giorgos Tolias, Yannis Avrithis, and Ondrej Chum. Label propagation for deep semi-supervised learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 5070–5079, 2019. 3
- [24] Yannis Kalantidis, Mert Bulent Sariyildiz, Noe Pion, Philippe Weinzaepfel, and Diane Larlus. Hard negative mixing for contrastive learning. *Advances in neural information processing systems*, 33:21798–21809, 2020. 1
- [25] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. *Advances in neural information processing systems*, 33:18661–18673, 2020. 3
- [26] Durk P Kingma, Shakir Mohamed, Danilo Jimenez Rezende, and Max Welling. Semi-supervised learning with deep generative models. *Advances in neural information processing systems*, 27, 2014. 1
- [27] Alex Krizhevsky and Geoffrey Hinton. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. 5
- [28] Svetlana Kutuzova, Oswin Krause, Douglas McCloskey, Mads Nielsen, and Christian Igel. Multimodal variational autoencoders for semi-supervised learning: In defense of product-of-experts. *arXiv preprint arXiv:2101.07240*, 2021. 1
- [29] Phuc H Le-Khac, Graham Healy, and Alan F Smeaton. Contrastive representation learning: A framework and review. *Ieee Access*, 8:193907–193934, 2020. 2
- [30] Dong-Hyun Lee et al. Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks. In *Workshop on challenges in representation learning, ICML*, page 896. Atlanta, 2013. 1
- [31] Kwonjoon Lee, Subhransu Maji, Avinash Ravichandran, and Stefano Soatto. Meta-learning with differentiable convex optimization. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10657–10665, 2019. 3
- [32] Hengduo Li, Zuxuan Wu, Abhinav Shrivastava, and Larry S Davis. Rethinking pseudo labels for semi-supervised object detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 1314–1322, 2022. 3
- [33] Lin Li, Jun Xiao, Hanrong Shi, Hanwang Zhang, Yi Yang, Wei Liu, and Long Chen. Nicest: Noisy label correction and training for robust scene graph generation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024. 3
- [34] Loren Lugosch, Tatiana Likhomanenko, Gabriel Synnaeve, and Ronan Collobert. Pseudo-labeling for massively multilingual speech recognition. In *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 7687–7691. IEEE, 2022. 3
- [35] Alex Nichol and John Schulman. Reptile: a scalable meta-learning algorithm. *arXiv preprint arXiv:1803.02999*, 2(3): 4, 2018. 3
- [36] Maria-Elena Nilsback and Andrew Zisserman. Automated flower classification over a large number of classes. In *Indian Conference on Computer Vision, Graphics and Image Processing*, 2008. 5
- [37] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, et al. Pytorch: An imperative style, high-performance deep learning library. *Advances in neural information processing systems*, 32, 2019. 5
- [38] Mary Phuong and Christoph Lampert. Towards understanding knowledge distillation. In *International conference on machine learning*, pages 5142–5151. PMLR, 2019. 3
- [39] Filip Radenovic, Abhimanyu Dubey, Abhishek Kadian, Todor Mihaylov, Simon Vandenhende, Yash Patel, Yi Wen, Vignesh Ramanathan, and Dhruv Mahajan. Filtering, distillation, and hard negatives for vision-language pre-training. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 6967–6977, 2023. 1
- [40] Alec Radford, Luke Metz, and Soumith Chintala. Unsupervised representation learning with deep convolutional generative adversarial networks. *arXiv preprint arXiv:1511.06434*, 2015. 2
- [41] Mamshad Nayeem Rizve, Kevin Duarte, Yogesh S Rawat, and Mubarak Shah. In defense of pseudo-labeling: An uncertainty-aware pseudo-label selection framework for semi-supervised learning. *arXiv preprint arXiv:2101.06329*, 2021. 3
- [42] Nikolaos Sarafianos, Xiang Xu, and Ioannis A Kakadiaris. Adversarial representation learning for text-to-image matching. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 5814–5824, 2019. 2
- [43] Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021. 2
- [44] Pierre Sermanet, Soumith Chintala, and Yann LeCun. Convolutional neural networks applied to house numbers digit classification. In *Proceedings of the 21st international conference on pattern recognition (ICPR2012)*, pages 3288–3291. IEEE, 2012. 5

- [45] Hardik Singh, Sweta Swagatika, Raavi Sai Venkat, and Sanjay Saxena. Justification of stl-10 dataset using a competent cnn model trained on cifar-10. In *2019 3rd International conference on Electronics, Communication and Aerospace Technology (ICECA)*, pages 1254–1257. IEEE, 2019. 5
- [46] Jake Snell, Kevin Swersky, and Richard Zemel. Prototypical networks for few-shot learning. *Advances in neural information processing systems*, 30, 2017. 3
- [47] Kihyuk Sohn, David Berthelot, Nicholas Carlini, Zizhao Zhang, Han Zhang, Colin A Raffel, Ekin Dogus Cubuk, Alexey Kurakin, and Chun-Liang Li. Fixmatch: Simplifying semi-supervised learning with consistency and confidence. *Advances in neural information processing systems*, 33:596–608, 2020. 6
- [48] Fariborz Taherkhani, Ali Dabouei, Sobhan Soleymani, Jeremy Dawson, and Nasser M Nasrabadi. Self-supervised wasserstein pseudo-labeling for semi-supervised image classification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12267–12277, 2021. 3
- [49] Yonglong Tian, Chen Sun, Ben Poole, Dilip Krishnan, Cordelia Schmid, and Phillip Isola. What makes for good views for contrastive learning? *Advances in neural information processing systems*, 33:6827–6839, 2020. 3
- [50] Chien-Chih Wang, Shaoyuan Xu, Jinmiao Fu, Yang Liu, and Bryan Wang. Kd-fixmatch: Knowledge distillation siamese neural networks. In *2023 IEEE International Conference on Image Processing (ICIP)*, pages 341–345. IEEE, 2023. 3
- [51] Xinjiang Wang, Xingyi Yang, Shilong Zhang, Yijiang Li, Litong Feng, Shijie Fang, Chengqi Lyu, Kai Chen, and Wayne Zhang. Consistent-teacher: Towards reducing inconsistent pseudo-targets in semi-supervised object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3240–3249, 2023. 3
- [52] Hao Wu and Saurabh Prasad. Semi-supervised deep learning using pseudo labels for hyperspectral image classification. *IEEE Transactions on Image Processing*, 27(3):1259–1270, 2017. 3
- [53] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017. 5
- [54] Qizhe Xie, Zihang Dai, Eduard Hovy, Thang Luong, and Quoc Le. Unsupervised data augmentation for consistency training. *Advances in neural information processing systems*, 33:6256–6268, 2020. 3
- [55] Qizhe Xie, Minh-Thang Luong, Eduard Hovy, and Quoc V Le. Self-training with noisy student improves imagenet classification. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10687–10698, 2020. 2
- [56] Pengxiang Yan, Guanbin Li, Yuan Xie, Zhen Li, Chuan Wang, Tianshui Chen, and Liang Lin. Semi-supervised video salient object detection using pseudo-labels. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 7284–7293, 2019. 3
- [57] Zhen Yang, Tinglin Huang, Ming Ding, Yuxiao Dong, Rex Ying, Yukuo Cen, Yangliao Geng, and Jie Tang. Batch-sampler: Sampling mini-batches for contrastive learning in vision, language, and graphs. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 3057–3069, 2023. 1
- [58] Bowen Zhang, Yidong Wang, Wenxin Hou, Hao Wu, Jindong Wang, Manabu Okumura, and Takahiro Shinozaki. Flexmatch: Boosting semi-supervised learning with curriculum pseudo labeling. *Advances in Neural Information Processing Systems*, 34:18408–18419, 2021. 3
- [59] Liheng Zhang, Guo-Jun Qi, Liqiang Wang, and Jiebo Luo. Aet vs. aed: Unsupervised representation learning by auto-encoding transformations rather than data. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2547–2555, 2019. 2
- [60] Zijun Zhang. Improved adam optimizer for deep neural networks. In *2018 IEEE/ACM 26th international symposium on quality of service (IWQoS)*, pages 1–2. Ieee, 2018. 6